

1 Relative motion

What students need to learn:

Relative motion of two particles, including relative displacement and relative velocity.

Problems may be set in vector form and may involve problems of interception or closest approach including the determination of course required for closest approach.

2 Elastic collisions in two dimensions

What students need to learn:

Oblique impact of smooth elastic spheres and a smooth sphere with a fixed surface.

3 Further motion of particles in one dimension

What students need to learn:

Resisted motion of a particle moving in a straight line.

The resisting forces may include the forms $a + bv$ and $a + bv^2$ where a and b are constants and v is the speed.

Damped and/or forced harmonic motion.

The damping to be proportional to the speed. Solution of the relevant differential equations will be expected.

4 Stability

What students need to learn:

Finding equilibrium positions of a system from consideration of its potential energy.

Positions of stable and unstable equilibrium of a system.

Unit M5 Mechanics 5

GCE Further Mathematics and GCE AS and GCE Further Mathematics (Additional)
A2 optional unit

M5.1 Unit description

Applications of vectors in mechanics; variable mass; moments of inertia of a rigid body; rotation of a rigid body about a fixed smooth axis.

M5.2 Assessment information

Prerequisites	A knowledge of the specifications for M1, M2, M3 and M4 and their prerequisites and associated formulae, together with a knowledge of scalar and vector products, and of differential equations as specified in FP2, is assumed and may be tested.
Examination	The examination will consist of one 1½ hour paper. It will contain about seven questions with varying mark allocations per question which will be stated on the paper. All questions may be attempted.
Calculators	Students are expected to have available a calculator with at least the following keys: +, −, ×, ÷, π , x^2 , $\sqrt{x}$, $\frac{1}{x}$, x^y , $\ln x$, e^x , sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

Formulae

Students are expected to know any other formulae which might be required by the specification and which are not included in the booklet, *Mathematical Formulae including Statistical Formulae and Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

Students will be expected to know and be able to recall and use the following formulae:

Work done by a constant force = $\mathbf{F} \cdot \mathbf{d}$

Angular momentum = $I\dot{\theta}$

Rotational kinetic energy = $\frac{1}{2}I\dot{\theta}^2$

$L = I\ddot{\theta}$

$$\int_{t_1}^{t_2} L \, dt = I\omega_2 - I\omega_1$$

For constant angular acceleration:

$$\omega_1 = \omega_0 + \alpha t$$

$$\omega_1^2 = \omega_0^2 + 2\alpha\theta$$

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\theta = \frac{(\omega_0 + \omega_1)}{2} t$$

1 Applications of vectors in mechanics

What students need to learn:

Solution of simple vector differential equations.

Vector differential equations such as

$$\frac{d\mathbf{v}}{dt} = k\mathbf{v},$$

$$\frac{d^2\mathbf{r}}{dt^2} + 2k\frac{d\mathbf{r}}{dt} + (k^2 + n^2)\mathbf{r} = \mathbf{f}(t)$$

where k and n are constants,

$$\frac{d\mathbf{r}}{dt} + 4\mathbf{r} = \mathbf{ie}'.$$

Use of an integrating factor may be required.

Work done by a constant force using a scalar product.

Moment of a force using a vector product.

The moment of a force $\mathbf{F}$ about O is defined as $\mathbf{r} \times \mathbf{F}$, where $\mathbf{r}$ is the position vector of the point of application of $\mathbf{F}$.

The analysis of simple systems of forces in three dimensions acting on a rigid body.

The reduction of a system of forces acting on a body to a single force, single couple or a couple and a force acting through a stated point.

2 Variable mass

What students need to learn:

Motion of a particle with varying mass.

Students may be required to derive an equation of motion from first principles by considering the change in momentum over a small time interval.